Prognosis of Uveal Melanoma in 500 Cases Using Genetic Testing of Fine-Needle Aspiration Biopsy Specimens

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Purpose: To determine the relationship between monosomy 3 and incidence of metastasis after genetic testing of uveal melanoma using fine-needle aspiration biopsy (FNAB).

Design: Noncomparative retrospective case series.

Participants: Five hundred patients.

Methods: Fine-needle aspiration biopsy was performed intraoperatively immediately before plaque radiotherapy. The specimen underwent genetic analysis using DNA amplification and microsatellite assay. Systemic follow-up was obtained regarding melanoma-related metastasis.

Main Outcome Measures: Presence of chromosome 3 monosomy (loss of heterozygosity) and occurrence of melanoma metastasis.

Results: Disomy 3 was found in 241 melanomas (48%), partial monosomy 3 was found in 133 melanomas (27%), and complete monosomy 3 was found in 126 melanomas (25%). The cumulative probability for metastasis by 3 years was 2.6% for disomy 3, 5.3% for partial monosomy 3 (equivocal monosomy 3), and 24.0% for complete monosomy 3. At 3 years, for tumors with disomy 3, the cumulative probability of metastasis was 0% for small (0–3 mm thickness), 1.4% for medium (3.1–8 mm thickness), and 23.1% for large (>8 mm thickness) melanomas. At 3 years, for tumors with partial monosomy 3, the cumulative probability of metastasis was 4.5% for small, 6.9% for medium, and [insufficient numbers] for large melanomas. At 3 years, for tumors with complete monosomy 3, the cumulative probability of metastasis was 0% for small, 24.4% for medium, and 57.5% for large melanomas. The most important factors predictive of partial or complete monosomy 3 included increasing tumor thickness ($P = 0.001$) and increasing distance to optic disc ($P = 0.002$).

Conclusions: According to FNAB results, patients with uveal melanoma demonstrating complete monosomy 3 have substantially poorer prognosis at 3 years than those with partial monosomy 3 or disomy 3. Patients with partial monosomy 3 do not significantly differ in outcome from those with disomy 3.

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Several articles have been written on the relevance of genetic testing of uveal melanoma.[1,2] In 1996, Prescher et al[3] reported the landmark observation that uveal melanoma with chromosome 3 monosomy was an important predictor of worse patient prognosis. In that report, the authors retrospectively evaluated 54 enucleated eyes with uveal melanoma and correlated the copy number of chromosome 3 to the known patient outcome. Several publications on genetic testing of melanoma from enucleated eyes have confirmed their observations.[4–8]

In 2002, Naus and associates[9] reported that fine-needle aspiration biopsy (FNAB) could be reliably used to sample tumors for genetic testing of uveal melanoma and the results correlated to those obtained by open biopsy after enucleation. In 2006, Midena and associates[10] reported the first clinical series of 8 eyes with uveal melanoma sampled by FNAB for genetic testing using fluorescent in situ hybridization. Later, Young et al[11] used transscleral FNAB into the tumor base for genetic testing in 18 eyes and found a 50% yield. Shields et al[12] added their experience with 140 cases of FNAB for uveal melanoma and indicated that trans pars plana FNAB into the tumor apex provided adequate DNA for microsatellite assay in 97% of cases with little complication and no patient with tumor recurrence along the needle tract. This report provides an analysis of patient prognosis in 500 cases based on results of genetic testing from FNAB for uveal melanoma.

Materials and Methods

We reviewed the clinical records of all patients on the Ocular Oncology Service at Wills Eye Institute, Philadelphia, Pennsylvania, with the diagnosis of uveal melanoma managed with FNAB yielding DNA for genetic testing of chromosome 3 status at the

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